

Zhengyu Zhuang

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Education

Boston University

09/2025-06/2027

M.S. in Electrical & Computer Engineering

Core Courses: MS Project, Deep Learning, Intro to Embedded System, Product Design

Rensselaer Polytechnic Institute

09/2021-06/2025

B.S. in Electrical Engineering

Research Projects

Autonomous Robotic System for Hydrogen Leak Detection

Industry-Sponsored Research Project (Plug Power)

Jan. 2025 – May. 2025

- Designed and implemented an autonomous ground rover integrating embedded control, obstacle avoidance, and UWB-based localization.
- Developed real-time sensing and data acquisition pipelines for hydrogen concentration monitoring (100–10,000 ppm) with low-latency wireless reporting.
- Achieved ± 5 cm positioning accuracy and 100% collision avoidance in field tests, improving hydrogen leak detection accuracy by approximately 35%.

EEG-Based Brain–Computer Interface Integration for Stroke Rehabilitation

Research Project & Technical Review

June. 2025 – Sep. 2025

- Conducted a comprehensive technical review of EEG-based brain–computer interface (BCI) systems for post-stroke motor rehabilitation.
- Analyzed the integration of BCIs with functional electrical stimulation (FES), immersive sensory technologies (AR/VR), and robotic rehabilitation systems from a system-level perspective.
- Compared rehabilitation effectiveness, neuroplasticity mechanisms, and human–machine interaction paradigms across different BCI-integrated approaches.
- Identified key challenges including EEG signal variability, real-time decoding latency, user fatigue, system calibration, and hardware cost, and discussed implications for practical clinical deployment.

Autonomous Maze-Solving Robot with Embedded Control and Path Planning

Capstone Design Project

Jan. 2025 – May. 2025

- Developed an autonomous maze-solving system using a TI-RSLK mobile robot with embedded control and real-time decision making.
- Implemented graph-based path planning using the A* algorithm to optimize navigation efficiency and minimize traversal time.
- Tuned PID control parameters through iterative experiments and LED-based testing to improve motion stability and path-following accuracy.
- Achieved approximately 10% reduction in average completion time through data-driven control optimization.

Automated Manufacturing System for Model Engine Assembly

Engineering Automation Project

Sep. 2024 – May. 2025

- Designed and implemented an automated manufacturing workflow integrating CNC machining, injection molding, and laser cutting for model engine production.
- Developed automation-oriented assembly processes to improve efficiency, consistency, and scalability.
- Coordinated system-level planning across fabrication, assembly, and quality control stages, emphasizing repeatability and process optimization.
- Delivered over 300 functional model engines within budget and schedule constraints, demonstrating

reliable end-to-end manufacturing automation.

Skills

Programming: C, C++, Python

Machine Learning & Deep Learning: Deep neural network design and training; optimization and regularization; convolutional, recurrent, and transformer-based models; representation learning; PyTorch

Embedded & Control Systems: Arduino, QEMU, Linux Device Drivers, PID Control

Manufacturing Design & Prototyping: CNC Machining, SolidWorks, Injection Molding

RESEARCH INTERESTS

- Autonomous and intelligent robotic systems with real-time perception, control, and decision-making
- Embedded and real-time systems for sensing, actuation, and safety-critical autonomy
- Human-machine interaction and shared control for autonomous platforms
- Brain-computer interfaces and neurotechnology for human-in-the-loop autonomous systems